Lesson 3
SOME USEFUL FUNCTIONS

Excel in Business

Text Functions

- LEFT
- RIGHT
- MID
- TRIM
- LEN

- FIND
- SEARCH
- CONCATENATE
- REPLACE

Let us understand the usage of these functions before we apply them to an actual data handling problem.

LEFT, RIGHT and MID

- LEFT extracts n characters from the left of a text string.
- RIGHT extracts n characters from the right of a text string.
- MID extracts n characters starting at position k in a text string.

LEFT, RIGHT and MID

SYNTAX

- LEFT(text, num of char)
- RIGHT(text, num of char)
- MID(text, starting position, num of char)

TRIM and LEN

- TRIM removes all spaces between words except for the single spaces. Thus if there were 3 spaces between the first and last name, then the trimmed result will have just one space.

  Syntax: TRIM(text)

- LEN returns the total number of characters in a text

  Syntax: LEN(text)

FIND and SEARCH

- FIND returns the start of the first occurrence of a string within a text. FIND is case sensitive.
- SEARCH works in a similar fashion to FIND except that it is **not** case sensitive.

Syntax of FIND and SEARCH

=FIND(text you want to locate,
where?,
start searching from what position?
)

SEARCH has the same syntax as FIND.

The CONCATENATE Function

CONCATENATE merges two or more strings into one.

Syntax: CONCATENATE(text1,text2,…)

In Cell D12 we use this function to join the strings stored in cells E2 and E3.

The REPLACE Function

Syntax:

=REPLACE(old text, k, m, new text)

REPLACE starts at position k and overwrites the next m characters of the old text with the new text.

Application of Text Functions

My data contains three fields: **product description**, **product ID**, and **product price.** Unfortunately, the worksheet I received had all three fields packed into one cell.

How do I put the fields in different columns in order to begin my analysis?

COUNT and COUNTA

- COUNT function counts the number of cells that contain numbers.

  Syntax: COUNT(range)

- COUNTA function counts the number of cells that are not empty.

  Syntax: COUNTA(range)

IF Function

IF function returns one value if a condition you specify evaluates to TRUE and another value if it evaluates to FALSE.

Syntax: IF(logical test, value(if true), value(if false))

Comparison Operators

Comparison operator	Meaning	Example
= (equal sign)	Equal to	A1=B1
> (greater than sign)	Greater than	A1>B1
< (less than sign)	Less than	A1<B1
>= (greater than or equal to sign)	Greater than or equal to	A1>=B1
<= (less than or equal to sign)	Less than or equal to	A1<=B1
<> (not equal to sign)	Not equal to	A1<>B1

Logical functions

- AND: Returns TRUE if all of its arguments are TRUE

  Syntax: AND(logical1, logical2,…)

- OR: Returns TRUE if any argument is TRUE

  Syntax: OR(logical1, logical2,…)

COUNTIF, COUNTIFS

COUNTIF function counts the number of cells within a range that meet the given criteria.

Syntax: COUNTIF(range, criterion)

COUNTIFS function counts the number of cells within a range that meet multiple criteria.

Syntax: COUNTIFS(range1, criterion1,
range2, criterion2,…)

SUMIF, SUMIFS

SUMIF function adds the cells within a range that meet the given criteria.

Syntax: SUMIF(criteria range, criteria, sum range)

SUMIFS function counts the number of cells within a range that meet multiple criteria.

Syntax: SUMIFS(sum range, criterion range1, criteria1,…)

NOTE: The order of arguments is different between SUMIFS and SUMIF

Database Functions

Name	Date	Product	Units	Dollars	Location
Betsy	4/1/2004	keyboard	45	$137.20	south
Hallagan	3/10/2004	pendrive	50	$152.01	midwest
Ashley	2/25/2005	headphone	9	$28.72	midwest
Hallagan	5/22/2006	keyboard	55	$167.08	west
Zaret	6/17/2004	keyboard	43	$130.60	midwest
Colleen	11/27/2005	mouse	58	$175.99	midwest
Cristina	3/21/2004	mouse	8	$25.80	midwest
Colleen	12/17/2006	keyboard	72	$217.84	midwest
Ashley	7/5/2006	mouse	75	$226.64	south
Betsy	8/7/2006	keyboard	24	$73.50	east
Ashley	11/29/2004	speaker	43	$130.84	east
Ashley	11/18/2004	keyboard	23	$71.03	west
Emilee	8/31/2005	keyboard	49	$149.59	west
Hallagan	1/1/2005	mouse	18	$56.47	south
Zaret	9/20/2006	pendrive	-8	$(21.99)	east
Emilee	4/12/2004	speaker	45	$137.39	east
Colleen	4/30/2006	speaker	66	$199.65	south
Jen	8/31/2005	keyboard	88	$265.19	midwest
Jen	10/27/2004	mouse	78	$236.15	south
Zaret	11/27/2005	keyboard	57	$173.12	midwest
Zaret	6/2/2006	speaker	12	$38.08	west

For this data, we want to know the following information:

Database Functions

1. How many dollars worth headphone did Jen sell?

2. How much revenue was generated in the east region or by Emilee?

3. What was the earning of Colleen or Zaret by selling headphone in the east region?

The DSUM Function

	A	B	C	D	E	F	G	H
1								
2								
3		Name	Date	Product	Units	Dollars	Location	
4		Betsy	4/1/2004	keyboard	45	$137.20	south	
5		Hallagan	3/10/2004	pendrive	50	$152.01	midwest	
6		Ashley	2/25/2005	headphone	9	$28.72	midwest	
7		Hallagan	5/22/2006	keyboard	55	$167.08	west	
8		Zaret	6/17/2004	keyboard	43	$130.60	midwest	
9		Colleen	11/27/2005	mouse	58	$175.99	midwest	
10		Cristina	3/21/2004	mouse	8	$25.80	midwest	
11		Colleen	12/17/2006	keyboard	72	$217.84	midwest	
12		Ashley	7/5/2006	mouse	75	$226.64	south	
13		Betsy	8/7/2006	keyboard	24	$73.50	east	
14		Ashley	11/29/2004	speaker	43	$130.84	east	
15		Ashley	11/18/2004	keyboard	23	$71.03	west	
16		Emilee	8/31/2005	keyboard	49	$149.59	west	
17		Hallagan	1/1/2005	mouse	18	$56.47	south	
18		Zaret	9/20/2006	pendrive	-8	$(21.99)	east	
19		Emilee	4/12/2004	speaker	45	$137.39	east	
20		Colleen	4/30/2006	speaker	66	$199.65	south	
21		Jen	8/31/2005	keyboard	88	$265.19	midwest	
22		Jen	10/27/2004	mouse	78	$236.15	south	
23		Zaret	11/27/2005	keyboard	57	$173.12	midwest	
24		Zaret	6/2/2006	speaker	12	$38.08	west	

=DSUM(data range,

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what column,

5

criteria for summation

)

criteria for summation is the range containing the information

Name	Product
Jen	keyboard

Points to Remember

- It is essential to include the exact column labels in the criteria range.
- Entries in the columns of the criteria range are ANDed.
- Entries in the rows of the criteria range are ORed. We will now see an example of this.

More than 1 row in criteria

	Name	Date	Product	Units	Dollars	Location
4	Betsy	4/1/2004	keyboard	45	$137.20	south
5	Hallagan	3/10/2004	pendrive	50	$152.01	midwest
6	Ashley	2/25/2005	headphone	9	$28.72	midwest
7	Hallagan	5/22/2006	keyboard	55	$167.08	west
8	Zaret	6/17/2004	keyboard	43	$130.60	midwest
9	Colleen	11/27/2005	mouse	58	$175.99	midwest
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15	Ashley	11/18/2004	keyboard	23	$71.03	west
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23	Zaret	11/27/2005	keyboard	57	$173.12	midwest
24	Zaret	6/2/2006	speaker	12	$38.08	west

criteria range is specified as

Name	Location
Emilee	
	east

to calculate the quantity:

earning of Emilee + earning in the east region

OO will not add Emilee's east earning twice

The DSUM Function

	Name	Date	Product	Units	Dollars	Location
4	Betsy	4/1/2004	keyboard	45	$137.20	south
5	Hallagan	3/10/2004	pendrive	50	$152.01	midwest
6	Ashley	2/25/2005	headphone	9	$28.72	midwest
7	Hallagan	5/22/2006	keyboard	55	$167.08	west
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What do you think the **criteria range**

Name	Location	Product
Colleen	east	headphone
Zaret	east	headphone

will calculate?

Database Functions

4. How many headphone transactions did not take place in the east?

5. How many dollars worth headphone did Jen sell during the year 2004?

6. How many units of accessories were sold for a price of at least $3.20?

The DCOUNT Function

	A	B	C	D	E	F	G	H
1								
2								
3		Name	Date	Product	Units	Dollars	Location	
4		Betsy	4/1/2004	keyboard	45	$137.20	south	
5		Hallagan	3/10/2004	pendrive	50	$152.01	midwest	
6		Ashley	2/25/2005	headphone	9	$28.72	midwest	
7		Hallagan	5/22/2006	keyboard	55	$167.08	west	
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21		Jen	8/31/2005	keyboard	88	$265.19	midwest	
22		Jen	10/27/2004	mouse	78	$236.15	south	
23		Zaret	11/27/2005	keyboard	57	$173.12	midwest	
24		Zaret	6/2/2006	speaker	12	$38.08	west	

=DCOUNT(data range,

B3:G27

what column,

any column that contains numeric entries

criteria range

criteria range is specified as

Location	Product
<>east	headphone

to count the number of transactions that did not take place in the east.

DATA TABLE

- Two-variable data tables use only one formula with two lists of input values. The formula must refer to two different input cells.

Location:

- DATA tab → DATA TOOL group → WHAT IF ANALYIS → DATA TABLE.

DATA TABLE

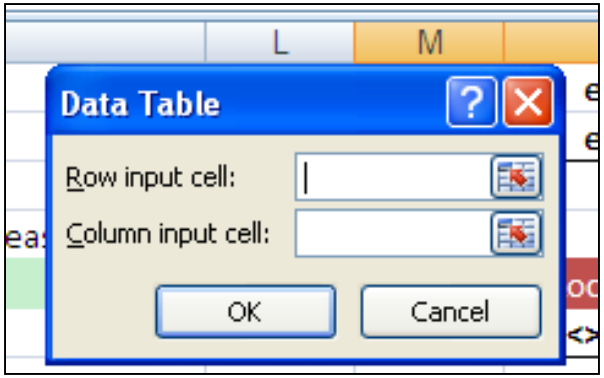
Refer to Workbook D functions:

- The formula corresponding to the example must be entered in the top left most cell e.g. in our case it is cell *M28*.

- After entering the formula, select the entire table, *M28:R37*

DATA TABLE

- Goto DATA TABLE



- In the Row input cell box, enter the reference to the input cell for the input values in the row.
- In the Column input cell box, enter the reference to the input cell for the input values in the column.